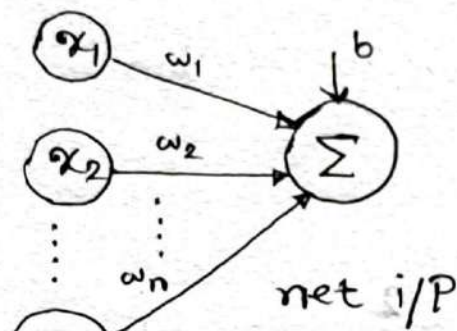


→ Activation Function in Neural Network

- weight
- adjustable weight
- tuning
- changable/fixed parameters
- bias value
- bias adjustable

i/p → input
 w → weight
 b → bias

"threshold value"
↳ 0



activation func. → $f(x_1w_1 + x_2w_2 + \dots + x_nw_n)$

Activation Function Result
(Linear Operation/Classification)
↳ [Boolean]

if $\boxed{\text{net} \geq 0}$ $y = 1$
else if $\boxed{\text{net} < 0}$ $y = 0$

"learning rate" → to change WEIGHT & BIAS
↳ α

↓ α : slow learn & converge
↑ α : fast learn
"trade off" & "balance"

{ Single Layer Architecture }

in supervised learning,
target value $\equiv$ predicted value
then → NO CHANGE

else, calculate ERROR

$\boxed{\text{error} = t - y}$
 $\quad = \text{target val} - \text{predicted val}$

and change WEIGHT w /
backward propagation

Shayan | ②

Activation Function

- ① Linear ($y=x$)
- ② Non-linear
- ③

✓ Single-Layer Perceptron
Multi-Layer Perceptron



Normal & Hidden
Layers of Neurons

Single Layer of
Neurons

Non-linear

- ① Sigmoid (S-curve) range [0-1]

$$A = \left(\frac{1}{1 + e^{-net}} \right)$$

Use case: Binary Classification

- ② Tanh range [-1 to 1]

$$A = \tanh(x) = \frac{2}{1 + e^{-2net}}$$

Use case:

- ③ ReLU range [0-∞]

$$A = \max(0, net)$$

- ④ Soft Plus

- ⑤ SoftMax (exponential curve)

Formula of:
Threshold Value

Deep Learning
↳ Observe

↳ Testing & Training
Accuracy

Shayan | ③

Q1: Design a Single Layer Perceptron to implement AND gate.
Initial Parameters:

Weights: $w_1 = 0.3$ $w_2 = 0.3$

Bias: $b = 0.3$

Learning rate: $\alpha = 0.2$

Threshold: $\theta = 1$

Use Activation Fn:

$$y = \begin{cases} 1 & \text{if net} \geq \theta \\ 0 & \text{if net} < \theta \end{cases}$$

গাঢ়ে dataset দেওয়া থাকবে না।

x_1	x_2	T
0	0	0
0	1	0
1	0	0
1	1	1

bias দেওয়া না থাকলে
bias calculate করা
লাগবে না

Step 1: Truth Table

Step 2: Perceptron Model

Net input:

$$\text{net} = x_1 w_1 + x_2 w_2 + b$$

Activation function:

$$y = \begin{cases} 1 & \text{if net} \geq \theta \\ 0 & \text{if net} < \theta \end{cases}$$

Error: $e = t - y$

Weight Update Rule: $w_i^{\text{new}} = w_i^{\text{old}} + \alpha \cdot e \cdot x_i$

Bias Update Rule: $b^{\text{new}} = b^{\text{old}} + \alpha \cdot e$

Input samples = 4

{ Run epochs until 4 input samples
are equal to the predicted input samples.

Epoch 1:

Pattern (0,0), $t=0$

$$\text{net} = 0 + 0 + 0.3 = 0.3 < 1$$

$$\Rightarrow y = 0$$

target, $t=0$

Since $t \equiv y$

No UPDATE needed.

Pattern (0,1), $t=0$

$$\text{net} = 0 + 0.3 + 0.3 = 0.6 < 1$$

$$\Rightarrow y = 0$$

target, $t=0$

Since $t \equiv y$

No Update needed.

Pattern (1,0), $t=0$

$$\text{net} = 0.3 + 0 + 0.3 = 0.6 < 1$$

$$\Rightarrow y = 0$$

target, $t=0$

Since $t \equiv y$

No Update needed.

hayan/⑤

Pattern (1,1), $t=1$

$$\text{net} = 0.3 + 0.3 + 0.3 = 0.9 < 1$$

$$\Rightarrow y = 0$$

Since $t \neq y$

Update needed.

$$w_1' = 0.3 + \alpha \cdot e \cdot x_1$$

$$= 0.3 + 0.2 \times 1 \times 1$$

$$= 0.5$$

$$w_2' = 0.3 + \alpha \cdot e \cdot x_2$$

$$= 0.3 + 0.2 \times 1 \times 1$$

$$= 0.5$$

$$b' = 0.3 + 0.2 \times 1 = 0.5$$

Epoch 2:

Pattern (0,0), $t=0$

$$\text{net} = 0 + 0 + 0.5 = 0.5 < 1$$

$$\Rightarrow y = 0$$

target, $t=0$

Since $t \equiv y$; No update.

Pattern (0,1), $t=0$

$$\text{net} = 0 \times 0.5 + 1 \times 0.5 + 0.5 = 1$$

$$\Rightarrow y = 1$$

$$t \neq y$$

$$e = t - y = -1$$

Update needed.

Shayan/6

$$w_1' = 0.5$$

$$w_2' = 0.3$$

$$b' = 0.3$$

Pattern (1,0), $t=0$

$$net = \dots$$

$$\Rightarrow y = 0$$

$$t \equiv y$$

No update.

Pattern (1,1), $t=1$

$$net =$$

$$\Rightarrow y = 1$$

$$t \equiv y$$

No update.

Step function

$$U(t) = \begin{cases} 1, & t \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

Epoch timeline

	0	0	0	1
E1	0	0	0	0
E2	0	1	0	1
E3	0	0	0	1

$$\begin{cases} w_1 = 0.5 \\ w_2 = 0.3 \\ b = 0.3 \end{cases}$$

Convergence Check

Input	Net	Output
(0,0)	0.3	0
(0,1)	0.6	0
(1,0)	0.8	0
(1,1)	1.1	1

Max Epoch = 5

Gradient Descent

Avoid

Q3. Dataset (Sigmoid Activation)

Question 2 epoch count 27773 978

Q4. Dataset (ReLU Activation)

$$\text{weight} \rightarrow w \alpha a \leftarrow \text{learning rate}$$

$$\alpha \rightarrow [0, 1]$$

never < 0

How α (learning rate) influences weights & bias update

#Biological & Artificial Neurons Comparison